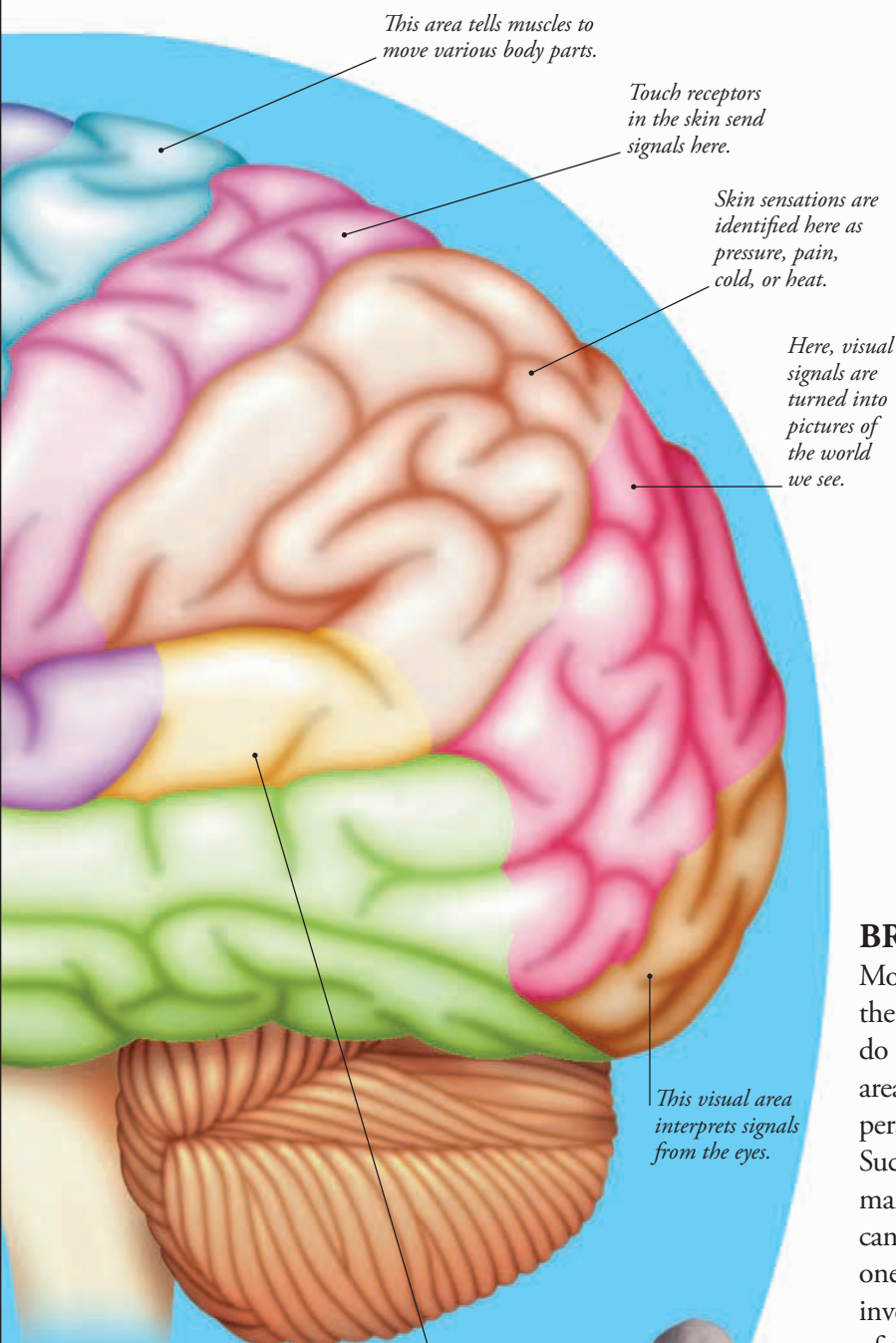
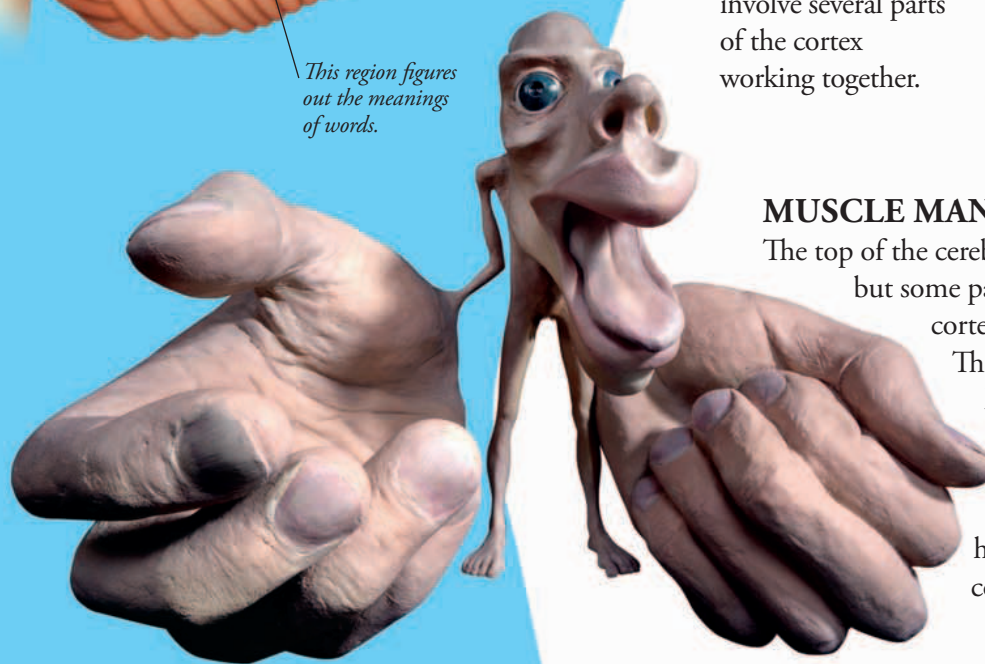


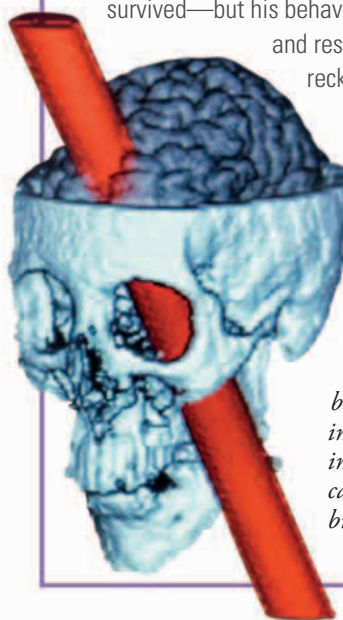


This region figures out the meanings of words.



A HOLE IN THE HEAD

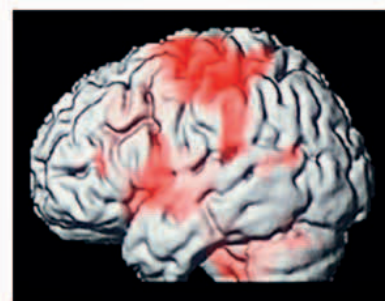
In September 1848, Phineas Gage, an American construction worker, met with a freak accident. An explosion propelled an iron rod through his head, damaging the front part of his brain. Incredibly, Gage survived—but his behavior changed from polite and responsible to rude and reckless. This led scientists to conclude that the front part of the brain controls personality.



◀ **GAGE'S BRAIN**
This computer image of Gage's skull and brain shows how he was injured. The rod went in under his left eye and came out through his brain and skull.

BRAIN IN ACTION

Modern scientists can use scanners to see which parts of the cerebral cortex “light up” when someone is asked to do something. For example, this scan shows several areas lighting up when a person moves a finger. Such studies show that many mental tasks can't be mapped to one site since they involve several parts of the cortex working together.



MUSCLE MAN

The top of the cerebral cortex controls muscles, but some parts of the body have a lot more cortex devoted to them than others.

This strange-looking model shows how much of the brain is involved in controlling different muscles. A large area of the cortex is devoted to moving the hands, but a much smaller area controls the arms and legs.